

Ryth Dasgupta

dasgupta.ryth@gmail.com

Systems Neuroscience, Computational Neuroscience, Multi-scale Modelling, Dynamical systems, AI/ML

Education

Université de Montréal, Canada

June 2025

MS Thesis in Computational Biology, Grade: 90.4/100

Indian Institute of Science Education and Research, Pune, India

June 2025

BS-MS in Biology, (Physics minor), CGPA: 8.6/10

Publications

1. Kale, Tanvi, **Ryth Dasgupta**, Mandar M. Inamdar, and Chaitanya A. Athale. "Mechanics of Escherichia coli cell width homeostasis and bulge dynamics from MreB and septum inhibition." *Molecular Biology of the Cell* 36, no. 10 (2025): ar126. <https://doi.org/10.1091/mbc.E24-12-0543>

Manuscript in preparation:

1. Exploring mechanisms for bacterial width change using coarse-grained simulations of the peptidoglycan cell wall, **Ryth Dasgupta**, Sven van Teeffelen.

Awards

- APS DBIO travel award - APS conference (2025)
- INSPIRE Scholarship (2021-2025)
- Mitacs Globalink Research Award (2024-2025)
- Gold Medal in the iGEM Competition, Paris (2022)
- Best Poster at Biophysics Paschim Conference, IIT Bombay (2024)
- Best Dry Lab, All India iGEM Meet (2022)
- Mitacs Globalink Fellowship (2023)
- AIC STeP Fellowship for Research Funding; iGEM (2022)

Conferences and Presentations

1. **Seminar Talk:** "Exploring mechanisms for bacterial width change using coarse-grained simulations of the peptidoglycan cell wall", Qbio.ca Canadian Biological physics seminar (UdeM Montréal, 2025)
2. **Poster:** "Emergence of E. coli cell shape from the mechanical balance between MreB bending rigidity and wall tension", Biophysics Pashchim Conference (IIT Bombay, 2024)
3. **Poster:** "Quantifying mechanical properties of bacterial cell wall", semester project (IISER Pune, 2023)
4. **Poster:** "Building a mathematical model to understand the dynamics of *E. coli* shape over a division cycle", semester project (IISER Pune, 2023)
5. **Poster:** "Mathematical model to optimise ACC deaminase production by *Azospirillum* to alleviate abiotic stress on plants", iGEM project (IISER Pune, 2022)

Research Experience

Bhalla Lab, NCBS

Project scientist
June 2026 – Present
Prof. Upinder S. Bhalla

Synaptic signalling in health, age and disease

- Agentic workflow to represent scientific literature in machine-readable schema
- Extending reaction networks in synaptic signalling
- Compare synaptic signalling between Young, Old and Diseased mice and humans

The Salt Inc., Mumbai

Founding team, Product lead
June 2025 – June 2026

MindLink: AI powered psychometric analysis for consumer insights

- Insight generation and display for dashboard, security measures and handling proprietary data
- Designed and built an agentic system for conversational insight retrieval

van Teeffelen Lab, University of Montreal

MS thesis student
May 2024 – May 2025
Prof. Sven van Teeffelen

Mechanism for bacterial width homeostasis using coarse-grained simulations of the cell wall

- Coarse-grained mechanistic model of bacterial cell wall remodelling
- Molecular dynamics simulations from scratch, high dimensional parameter optimization and paper writing

CyCelS Lab, IISER Pune

Undergraduate Researcher
Jan 2023 - May 2024
Prof. Chaitanya Athale

Biophysical modelling of the mechanics of cell in *E.coli*

- Developed a dynamical systems model to study the regulation of *E. coli* cell width and length
- Characterised the mechanical properties of the *E.coli* cell envelope using global optimisation algorithms to match biophysical model to experimental data
- Co-wrote a [paper](#) as the second author

Layton Lab, University of Waterloo

Mitacs Globalink Research Fellow
May 2023 – Aug 2023
Prof. Anita Layton

Modeling fuel homeostasis during exercise in aging men

- Developed a multi-scale ODE framework for modeling fuel homeostasis during exercise
- Performed parameter estimation and sensitivity analysis using global optimization methods
- Simulated age-dependent metabolic regulation and found key metabolic dysregulations that correlate with age

International Genetically Engineered Machines (iGEM)

Dry Lab Lead
Jan 2022 – Oct 2022
Prof. M.S. Madhusudhan

iGEM Synthetic Biology Project

- Modelling bacterial metabolism using flux balance analysis and diffusion models
- Sensitivity analysis and parameter optimization for kinetic models
- Managed a team of 5 people, coordinating the conceptualization, progress and [documentation](#) of research